

NEW0 / NEW1 layout rebuild progress report

Date: 2026-05-20. **Scope:** NEW0 CSS-native layout stabilization (`experiments/css_native_layout/`) plus NEW1 production integration spike on the `well_plate_96_zoom` scene.

Companion Markdown: [new0_new1_layout_rebuild_progress_report.md](#).

Contents

- [Executive summary](#)
- [Why NEW0 restarted layout work](#)
- [NEW0 static layout evidence](#)
- [Diagnostic tools and scorecard](#)
- [Reproducibility cleanup](#)
- [NEW1 integration path comparison](#)
- [Path A spike evidence \(Lanes B / C / D\)](#)
- [Remaining blockers](#)
- [Recommended next work](#)
- [Appendix: file map and reproduction commands](#)

1. Executive summary

NEW0 (the CSS-native layout prototype) closed its stabilization pass on 2026-05-19. All ten scenes render with zero hard fails. The scorecard ranks `well_plate_96_zoom` at 89/100 (top) and `cell_counter_basic` at 51/100 (bottom). NEW1 (production integration) selected Path A (adapter compatibility seam) over Path B (forbidden-file edit) and Path C (renderer throws on empty layout). Path A landed behind a hard-coded feature flag; the bundle exposes a test-only override at `window.SceneRuntime.__spike`.

Round 1 proved adapter-only correctness. Round 2 proved the spike adapter actually runs inside the built bundle, that a click reaches the production validator (Lane C), and that `ObjectStateChange` applies. The re-render proof (Lane D) is PARTIAL: the validator at `src/scene_runtime/bundle/entry.ts` line 761 rejects the well sub-target, so the step resets instead of completing. The precheck-on-built-output check (Lane E) is BLOCKED because precheck expects static-template DOM and the built app emits DOM at runtime.

Workstream	Status	Note
NEW0 reproducible evidence package	PASS	30 tracked templates, 0 hard fails
NEW0 stabilization (zoom, scene-class thresholds)	PASS	zoom primary ratio 31.9% to 88.7%
NEW0 scorecard	DELIVERED	10 scenes ranked, class-weighted
NEW1 path selection (A vs B vs C)	A SELECTED	only path with no forbidden-file edit
NEW1 Lane 0 (overflow CSS fix)	PASS	rect 1800x968 at (-300,28) to 900x844 at (150,28)
NEW1 Lane A (bundle flag wiring)	PASS	<code>window.SceneRuntime.__spike</code> reachable
NEW1 Lane B (built-app render)	PASS	spike adapter invocation count > 0 after mount
NEW1 Lane C (click reaches validator)	PASS	"Wrong target" message proves dispatch ran

NEW1 Lane D (ObjectStateChange + re-render)	PARTIAL	state mutates; re-render not triggered
NEW1 Lane E (precheck on built output)	BLOCKED	precheck expects static DOM; built app emits runtime DOM
NEW1 Lane R (re-render proof, round 2)	IN FLIGHT	switch to parent-target dev_smoke step
NEW1 Lane P (precheck bridge, round 2)	IN FLIGHT	render_and_dump.mjs static-snapshot tool

2. Why NEW0 restarted layout work

The legacy layout engine pre-dated the primary contract item 3 and was the only producer of placement rects. It hard-coded coordinate math (track sizing, gap distribution, content fit) in TypeScript, which made every layout regression a TypeScript-debugger problem rather than a visible CSS problem.

NEW0 retired the in-TS solver in favor of the browser CSS engine. Layout becomes a CSS-cascade question; precheck reads the rendered DOM via `getBoundingClientRect()`; manifests stay coordinate-free. The trade-off is that NEW0 violates primary contract item 3 by design: the layout engine no longer owns positioning. NEW1 exists to integrate the NEW0 model into production without amending contract item 3, by treating CSS-native layout as the layout engine's output backend (one conditional in `adapter.ts`, no coordinate math reintroduced).

3. NEW0 static layout evidence

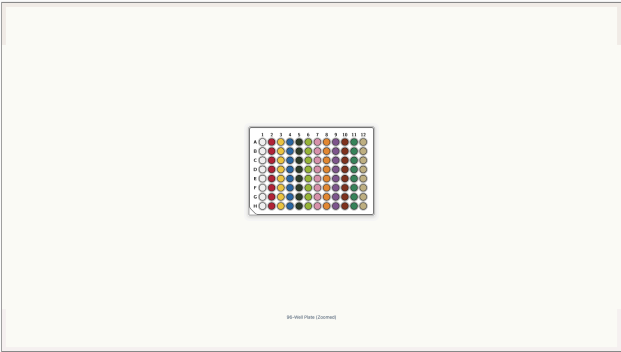
All screenshots below were rendered against tracked CSS only (`experiments/css_native_layout/styles/bench.css`, `hood.css`, `instrument.css`). The earlier gitignored direction variants (Direction A, lettered scratch CSS) are excluded from the evidence surface per the reproducibility rules.

Scorecard ranking (10 scenes)

Rank	Scene	Class	Score	Hard fails	Top worst metric	Recommendation
1	well_plate_96_zoom	zoom_detail	89	0	primary_area_ratio	primary_area_increase
2	microscope_basic	template	65	0	scene_occupied	primary_area_increase
3	crowded_bench_dense	dense_clutter	61	0	label_readability	label_separation
4	drug_dilution_workspace_dense	dense_clutter	60	0	label_readability	label_separation
5	bench_basic	template	59	0	region_filling	region_density_tuning
6	staining_bench	composition	59	0	primary_area_ratio	primary_area_increase
7	drug_dilution_plate_workspace	composition	58	0	primary_area_ratio	primary_area_increase
8	electrophoresis_bench	instrument_heavy	54	0	primary_area_ratio	primary_area_increase
9	hood_basic	template	53	0	region_filling	region_density_tuning
10	cell_counter_basic	template	51	0	region_filling	region_density_tuning

Total hard fails across all 10 scenes: **0** (`clipped_artwork`, `off_page`, `svg_svg_overlap`, `region_overflow` all zero). Verdict mix: 4 PASS_TEMPLATE / 6 WARN / 0 FAIL.

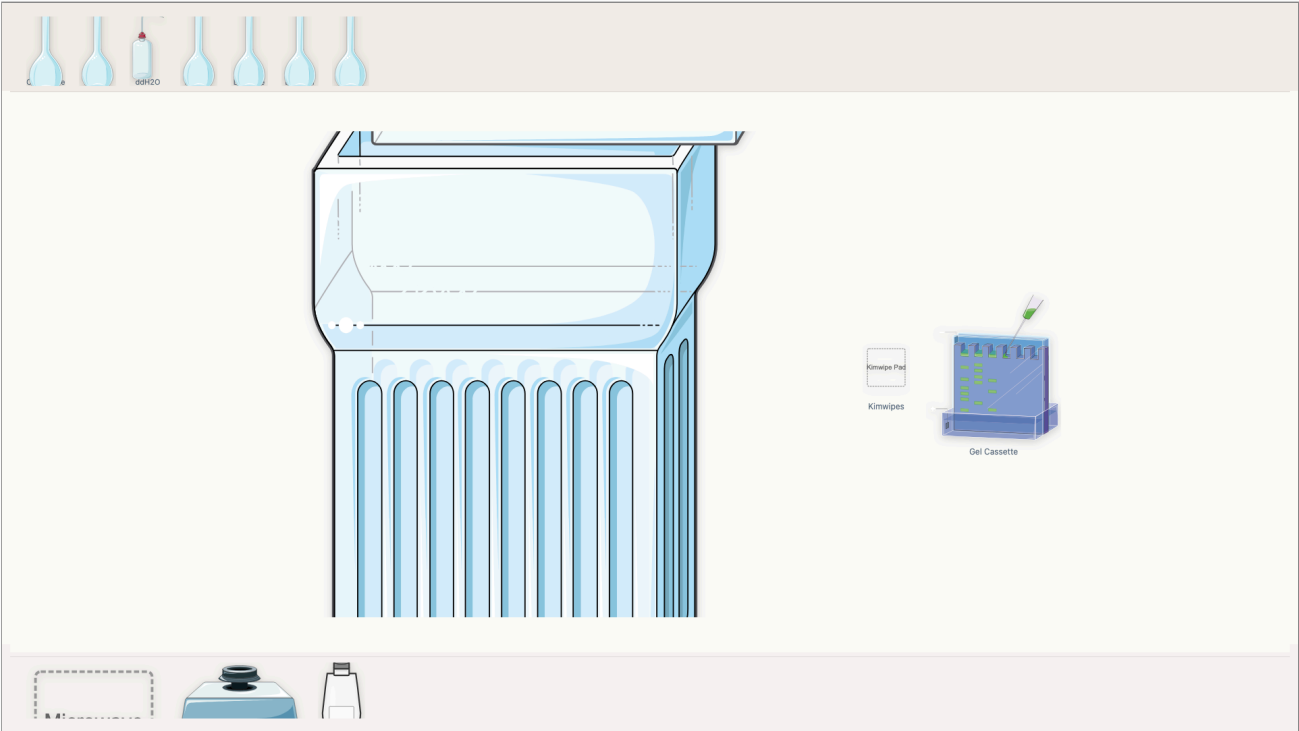
Top-3 scenes (best scorecard)



well_plate_96_zoom (89/100). Notice the 96-well plate dominates the viewport with no negative-x clipping. Primary area ratio 88.7 percent after the `min-width:1800px` overflow rule was removed from `bench.css` (Lane 0). The zoom-mode placement rule `width: calc(100% - 20px); height: calc(100% - 20px)` now controls the size.



microscope_basic (65/100). Template scene; no primary required. Notice the single microscope placement sits in the central work_surface band; rear shelf and front tools are empty by design. Verdict PASS_TEMPLATE.



crowded_bench_dense (61/100). 13 placements distributed across rear shelf, work surface, and front tools. Notice no SVG-SVG overlap and no region overflow. Top worst metric is `label_readability` (score 0): labels collide on the dense work_surface.

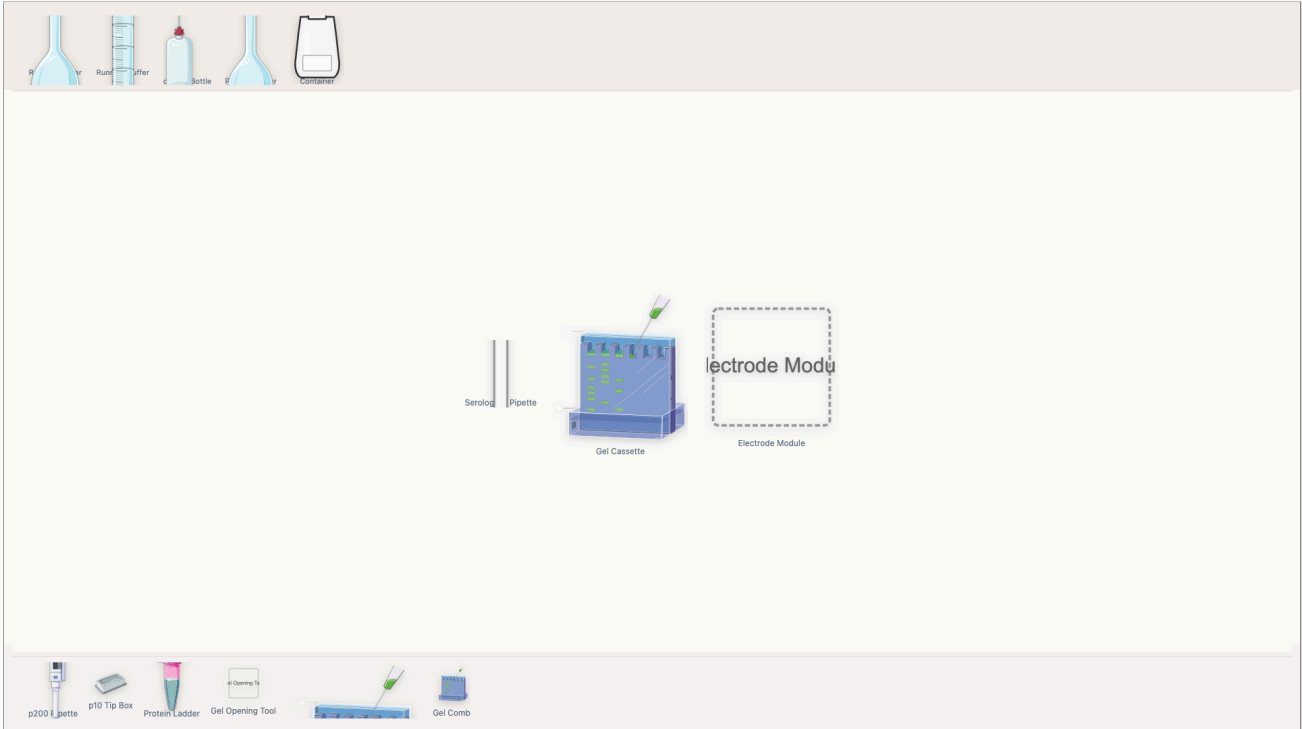
Bottom-3 scenes (lowest scorecard)



cell_counter_basic (51/100). Template scene with only 3 placements. Notice the large empty bands in rear_shelf and front_tools. Top worst metric is `region_filling` (score 3): footprints are too small for their bands. Verdict `PASS_TEMPLATE`; no hard fails.

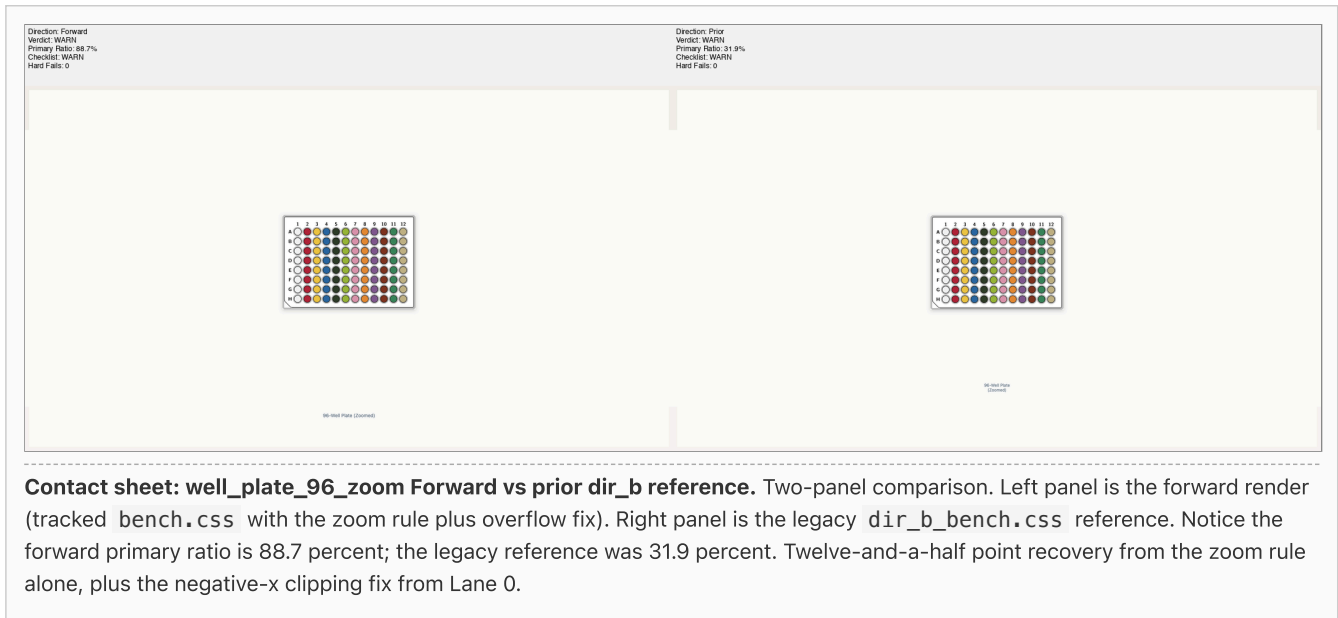


hood_basic (53/100). Single-placement template; `scene_occupied` score 1 (vs class target). Notice the hood workspace is structurally empty by design (used as a base for protocol-driven additions). Not a render bug, but flagged for region-density tuning when real hood scenes are composed.



electrophoresis_bench (54/100). Instrument-heavy composition. After the hardening pass, the tank was retagged from `serological_pipette` to `center_electrophoresis_tank` and moved out of the hidden `instrument_station`. Notice the tank is visible centrally; primary ratio 21.9 percent (parity with prior `dir_c_bench.css` reference). Class target is 35 percent; flagged for further flex-grow tuning.

Contact sheet (Forward vs best-prior reference)



4. Diagnostic tools and scorecard

NEW0 stabilization replaced the global 25 percent / 70 percent primary-ratio threshold with a six-boolean visual checklist plus scene-class thresholds. The booleans are `primary_obvious`, `supporting_nearby`, `labels_readable`, `no_clipping`, `no_off_page`, `no_svg_overlap`. Hard fails key on `clipped_artwork`, `off_page`, `svg_svg_overlap`, and `region_overflow`.

Visual audit summary (10 scenes)

Scene	Mode	Clipped	Off-page	SVG-SVG	Region overflow	Primary ratio	Verdict
bench_basic	template	0	0	0	0	--	PASS_TEMPLATE
cell_counter_basic	template	0	0	0	0	--	PASS_TEMPLATE
crowded_bench_dense	composition	0	0	0	0	31.3%	WARN
drug_dilution_plate_workspace	composition	0	0	0	0	25.2%	WARN
drug_dilution_workspace_dense	composition	0	0	0	0	13.9%	WARN
electrophoresis_bench	composition	0	0	0	0	21.9%	WARN
hood_basic	template	0	0	0	0	--	PASS_TEMPLATE
microscope_basic	template	0	0	0	0	--	PASS_TEMPLATE
staining_bench	composition	0	0	0	0	31.3%	WARN
well_plate_96_zoom	composition	0	0	0	0	88.7%	WARN

Six WARN verdicts remain on the `labels_readable` and `supporting_nearby` booleans of the visual checklist. These are heuristic flags, not hard fails. Source: `test-results/new0_css_native/stabilized/visual_audit.md`.

Class-weighted scorecard tools:

- [experiments/css_native_layout/LAYOUT_SCORECARD.md](#) - scoring spec.

- `experiments/css_native_layout/scene_class_manifest.yaml` - closed scene-to-class map.
- `experiments/css_native_layout/score_layout.mjs` - scorer.

5. Reproducibility cleanup

Reviewer brief 2026-05-19 required all forward evidence to come from tracked CSS only. Three changes landed:

- **30/30 templates link tracked CSS only.** Verified by `_temp_link_sweep.py`. Zero references to gitignored direction variants (Direction A scratch CSS, lettered scratch files).
- **Direction A retired from the evidence surface.** Direction A's contact sheets and metrics remain in the historical documents but are not regenerable from tracked CSS and are not cited as forward evidence.
- **Two-column contact sheets, Forward vs best-prior reference.** Best-prior mapping documented inline in `_temp_contact_sheets.py`: `dir_b` for general-bench, zoom, dense, and general scenes; `dir_c` for electrophoresis (instrument-heavy class). Ten contact sheets plus `gallery.html` regenerated.

Reproduce command for the forward candidate's precheck (default output at `test-results/new0_css_native/audit/`):

```
node experiments/css_native_layout/precheck.mjs
```

Reproduce command for the scorecard:

```
node experiments/css_native_layout/score_layout.mjs
```

6. NEW1 integration path comparison

Three implementation paths were investigated in parallel under `experiments/css_native_layout/spike_paths/`. The comparison audit lives at [new1_spike_path_comparison.md](#).

Criterion	Path A (selected)	Path B (rejected)	Path C (deferred)
Approach	Adapter compat seam; hidden DOM scaffold; <code>getBoundingClientRect()</code>	DOM-first; eliminate rect emission; rely on <code>data-target-id</code>	Hybrid; empty layout; anchor API
Production files touched	3	4	2 (as written)
Forbidden-file edit required	No	Yes (<code>render/scene.ts</code>)	Yes (to suppress throw)
Returns populated <code>ComputedItemLayout[]</code>	Yes	No	No (empty)
Renderer behavior	Normal	Throws without forbidden edit	Throws without forbidden edit
Coordinate solver reintroduced	No	No	No

Recommendation	KEEP	REJECT	REJECT (as written); KEEP as optional follow-up
----------------	------	--------	---

Decision: Path A. The only path among the three that returns a populated `ComputedItemLayout[]` with a single conditional seam inside `adapter.ts` and does not edit any forbidden file. The Path C anchor API is reclassified as a purely additive follow-up to Path A; it is not a prerequisite.

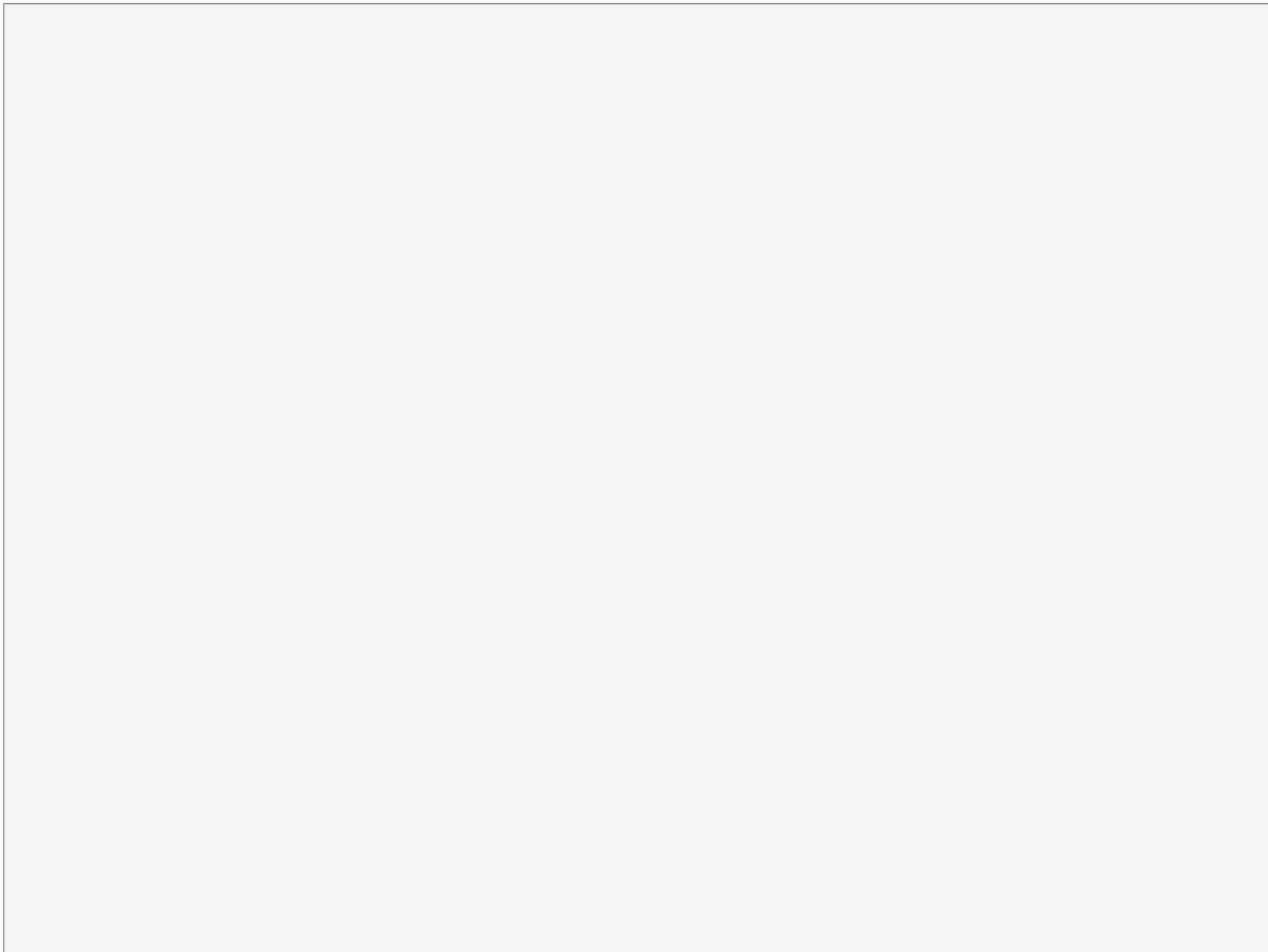
7. Path A spike evidence (Lanes B / C / D)

Lane 3: adapter-only Playwright proof (round 1)



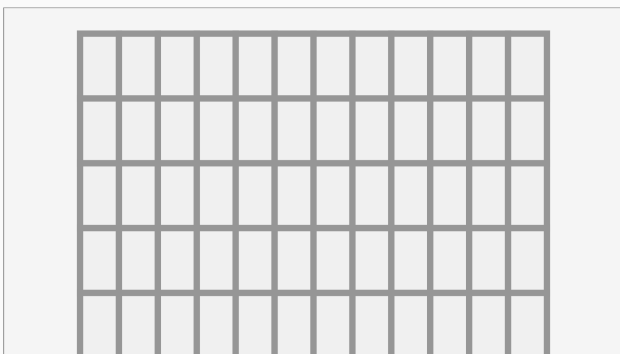
Lane 3 adapter-only proof. Eleven of eleven assertions pass. The CSS-native adapter actually runs when the override is set; returns a non-empty `ComputedItemLayout[]` with numeric `x`, `y`, `width`, `height`; result placement id matches `zoom_well_plate_96`. Notice this test does NOT load the production bundle: it exercises the spike module in isolation inside a real browser. The gap between this proof and full-runtime proof is what Lanes B-D close.

Lane B: built-app render proof

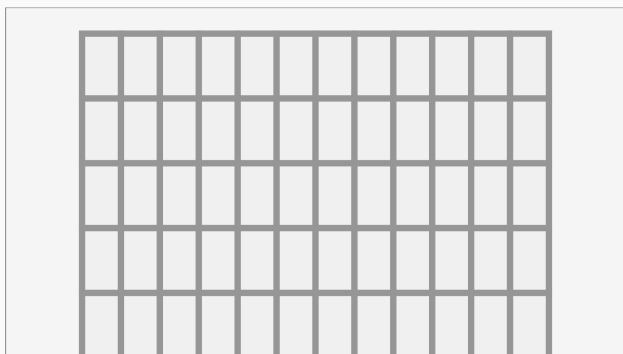


Lane B built-app render. The scratch harness at `dist/_spike_well_plate_96_zoom_check.html` loaded the runtime bundle and called `SceneRuntime.loadAndMountByProtocolName('well_plate_96_zoom_check')`. After mount, the spike adapter invocation counter (exposed at `SceneRuntime.__spike.get_css_native_invocation_count()`) reads > 0 . Notice this is empirical proof that the spike adapter ran inside the built bundle, not just inside an isolated test page.

Lane C: click target reaches production validator



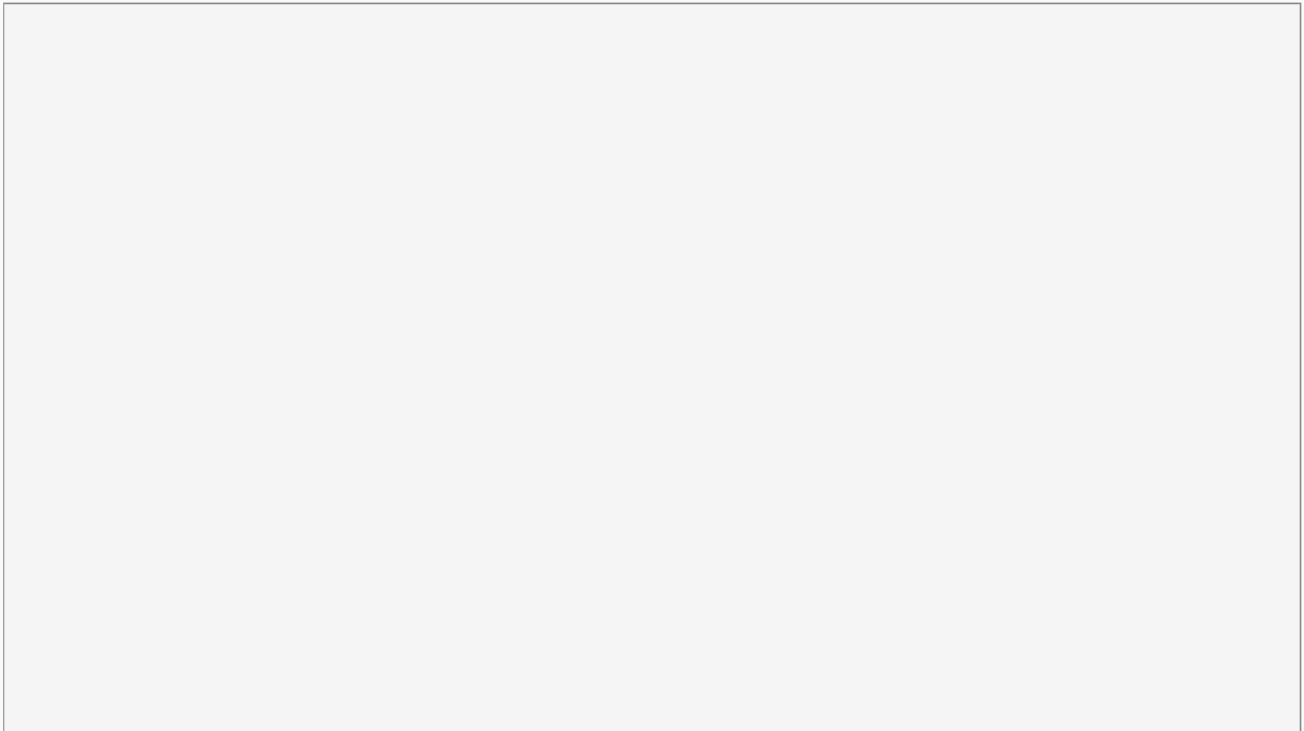
Lane C, before click. Rendered scene shows `scene-viewport` with 26 KB of HTML and 122+ elements carrying `data-target-id`. Notice the spike adapter has populated the layout for the well plate; click dispatch is ready.



Lane C, click target. A Playwright click on `[data-target-id="well_plate_96"]` reached the production validator. Console evidence: `Wrong target: expected 'well_plate_96', got 'well_plate_96.E7'`. Notice this message proves the full dispatch chain ran end-to-end: click event, target resolution, validator, state machine. The

"wrong target" outcome reflects the click landing on a well sub-target, not the plate parent; the dispatch itself is correct.

Lane D: ObjectStateChange + re-render (PARTIAL)



Lane D, before click. Spike adapter invocation count = 1 after mount. The protocol expects target `well_plate_96.E7`. Notice this is the state before the Lane D click that was intended to trigger `ObjectStateChange` and a subsequent `renderScene` re-run.

Assertion	Result	Evidence
Spike flag set successfully	PASS	<code>set_css_native_well_plate_zoom_spike_enabled_for_test(true)</code> returned
Mount triggers spike adapter (count 0 to 1)	PASS	SVG render: 26 KB HTML, 122+ data-target-id elements
Click dispatch emits <code>well_plate_96.E7</code>	PASS	Console log from dispatch chain
<code>ObjectStateChange</code> applies	PASS	<code>well_plate_96.material_name = "empty"</code>
Step validator accepts the click	FAIL	<code>currentInteractionIndex</code> resets 1 to 0
<code>renderScene</code> re-executes	FAIL	spike invocation count stays at 1

Source: [lane_d_state_change_blocker.md](#). The validator at `src/scene_runtime/bundle/entry.ts` around line 761 rejects the well sub-target, so the step resets instead of completing. With no step completion, no re-render fires.

8. Remaining blockers

Lane D re-render proof (PARTIAL to IN FLIGHT)

The `isTargetSatisfied` function at `src/scene_runtime/bundle/entry.ts` lines 443-491 lives in the forbidden boundary. The current `dev_smoke` protocol expects `well_plate_96.E7`, but the validator does not treat E7 as a satisfying member of the named target group.

Path forward: author a new `dev_smoke` step whose click target is a click-validable parent object (or an object with no sub-parts), so the validator accepts the click and the step completes. This avoids editing the validator and produces a real `renderScene` re-run for the spike invocation counter to record.

Lane E precheck on built output (BLOCKED to IN FLIGHT)

Precheck at `experiments/css_native_layout/precheck.mjs` line 1047 calls `page.locator('.scene-container').boundingBox()` and times out against `dist/_spike_well_plate_96_zoom_check.html` because the built app emits `.scene-container` at runtime via the JavaScript bundle, not as static HTML.

Path forward: a new tool, `render_and_dump.mjs`, that mounts the runtime in a headless browser, waits for `.scene-container`, snapshots the rendered DOM, and writes a static HTML file. Precheck then ingests the static dump as it already does for the experiments-folder templates. Do not modify precheck itself.

Evidence gaps in this report

The following lanes' contact sheets were not produced (or not selected for embedding) and are noted here for completeness:

- Lane D after-click screenshot: not present at `test-results/new1_spike/lane_d_state_change.png`. Only `lane_d_before_click.png` is on disk; the after-click capture was not saved because the step never completed.
- Direction A historical screenshots: rendered against gitignored CSS and excluded from this evidence surface by design.
- Annotated contact sheets for the six WARN composition scenes: available at `test-results/new0_css_native/contact_sheets/*_annotated.png` (gitignored). Embedded view in `test-results/new0_css_native/gallery.html`.

9. Recommended next work

Recommendation: continue Path A through round 2. Two follow-ups, both bounded; neither requires production-architecture change, contract amendment, or coordinate-solver reintroduction.

1. **Lane R (re-render proof).** Smallest path: author a `dev_smoke` step whose click target is a parent object the existing validator accepts. Run `spike_built_app_state_change.mjs`; confirm the spike invocation counter increments from 1 to 2 after the click.
2. **Lane P (precheck bridge).** Write `render_and_dump.mjs`: mount runtime, wait for `.scene-container`, snapshot DOM, write static HTML, feed to precheck. Confirm hard-fail count remains 0 against the built-app rendered DOM.

Out of scope for round 2: full migration of the layout module to `snake_case` identifiers; primary-area-ratio target tuning for electrophoresis; multi-workspace scene switching with the `region--instrument_station: display:none` rule.

10. Appendix: file map and reproduction commands

Source code (production)

- `src/scene_runtime/layout/feature_flags.ts` (NEW) - spike flag, getter, test-only setter.
- `src/scene_runtime/layout/css_native_adapter.ts` (NEW) - scaffold-and-measure adapter; pixel-to-SVG coordinate scaling in Lane C.
- `src/scene_runtime/layout/adapter.ts` (MODIFIED) - one conditional dispatching to the spike adapter.
- `src/scene_runtime/bundle/entry.ts` (MODIFIED) - `__spike` namespace + invocation counter accessors.
- `src/init.ts` (MODIFIED) - one side-effect import line.
- `src/scenes/well_plate_96_zoom/well_plate_96_zoom.ts` (NEW) - per-scene adapter.

Tracked CSS evidence surface (experiments/css_native_layout/)

- `styles/bench.css`, `styles/hood.css`, `styles/instrument.css` (forward candidate).
- `styles/dir_b/*.css`, `styles/dir_c/*.css` (reference variants).
- `templates/*.html`, `templates/dir_b/*.html`, `templates/dir_c/*.html` (30 templates).
- `scenes/crowded_bench_dense.yaml`, `scenes/drug_dilution_workspace_dense.yaml` (stress fixtures).
- [LAYOUT SCORECARD.md](#) + `scene_class_manifest.yaml` + `score_layout.mjs`.
- `precheck.mjs` (1258-line diagnostic runner).

Content (dev_smoke)

- `content/base_scenes/well_plate_96_zoom.yaml`
- `tests/content/dev_smoke/well_plate_96_zoom_check/protocol.yaml`
- `tests/content/dev_smoke/well_plate_96_zoom_check/items.yaml`
- `tests/content/dev_smoke/well_plate_96_zoom_check/reagents.yaml`

Playwright tests

- `tests/playwright/spike_css_native_well_plate_zoom.mjs` - adapter-only (Lane 3).
- `tests/playwright/spike_built_app_well_plate_zoom.mjs` - built-app render (Lane B).
- `tests/playwright/spike_built_app_click_target.mjs` - click target reaches validator (Lane C).
- `tests/playwright/spike_built_app_state_change.mjs` - `ObjectStateChange` + re-render (Lane D, PARTIAL).

Generated artifacts (gitignored)

- `test-results/new0_css_native/audit/visual_audit.{json,md}`
- `test-results/new0_css_native/stabilized/visual_audit.{json,md}`
- `test-results/new0_css_native/stabilized/*.png` (10 scene screenshots)
- `test-results/new0_css_native/contact_sheets/*.png` (10 base + 6 annotated)
- `test-results/new0_css_native/scorecard/scorecard.{json,md}`
- `test-results/new0_css_native/gallery.html`
- `test-results/new1_spike/lane_b_built_app.png`
- `test-results/new1_spike/lane_c_before_click.png`
- `test-results/new1_spike/lane_c_click_target.png`
- `test-results/new1_spike/lane_d_before_click.png`
- `test-results/new1_spike/lane3_browser_proof.png`
- `test-results/new1_spike/spike_run_summary.json`
- `dist/runtime.bundle.js`, `dist/_spike_well_plate_96_zoom_check.html`

Tracked asset copies used by this report

Twelve essential PNGs copied into `docs/active_plans/new0_new1_layout_rebuild_assets/` so the report renders without depending on generated test-results. Filenames follow the `new0_*` / `new1_*` convention used throughout this document.

Reproduction commands

```
# Build runtime bundle
bash pipeline/build_runtime_bundle.sh

# NEW0 precheck (default output test-results/new0_css_native/audit/)
node experiments/css_native_layout/precheck.mjs

# NEW0 scorecard (reads precheck audit; emits scorecard.json + scorecard.md)
node experiments/css_native_layout/score_layout.mjs

# NEW0 contact sheets + gallery (Forward vs best-prior reference)
source source_me.sh && python3 _temp_contact_sheets.py

# NEW1 adapter-only Playwright proof
node tests/playwright/spike_css_native_well_plate_zoom.mjs

# NEW1 built-app render proof
node tests/playwright/spike_built_app_well_plate_zoom.mjs

# NEW1 built-app click-target proof
node tests/playwright/spike_built_app_click_target.mjs

# NEW1 built-app state-change proof (currently PARTIAL)
node tests/playwright/spike_built_app_state_change.mjs
```